

Diffusion, Gelation, Spherification

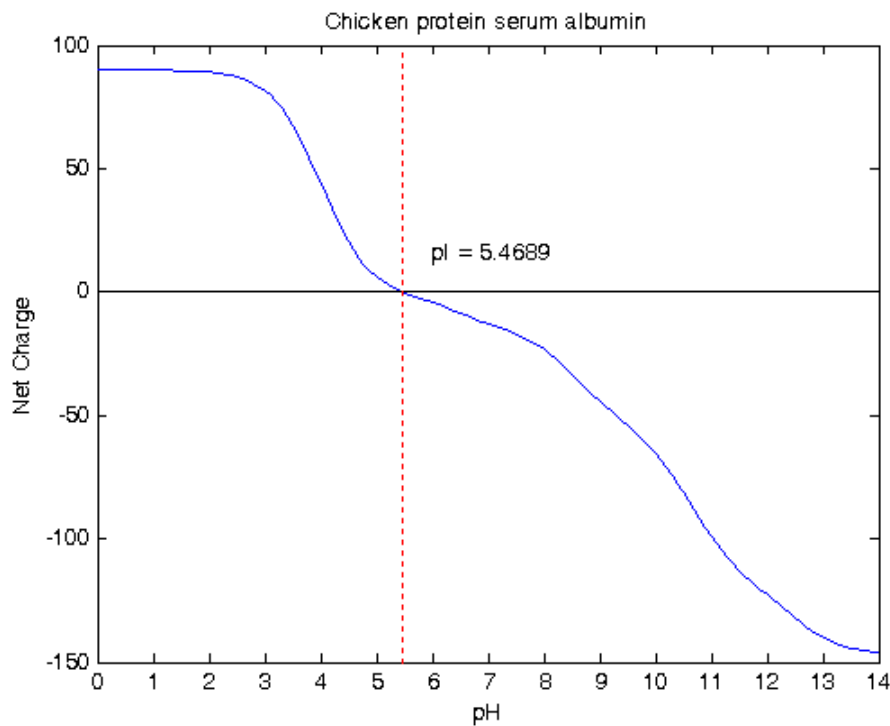
1. How many seconds would it take for a molecule with a diffusion constant of $3 \cdot 10^{-6} \text{ cm}^2/\text{s}$ to diffuse 1 mm?

2. You put a raw egg in a vinegar solution with pH 5.

A. Based on the diagram below, what is the net charge of the protein at neutral pH, (i.e. pH 7)?

B. What is the net charge at pH 5? Why did it change?

C. What do you expect to observe when you crack open the egg after marinating in this solution for 1 week? What about if the pH is 3?



Heat Transfer

1. Chef Corey seared piece of tuna. It turns out great: perfect temperature at the center and a nice brown crust. Draw a diagram of the temperature in the fish as a function of distance from the center *right after you take* it off the griddle.
2. Explain why it is hard to cook a piece of tuna (or steak).
3. The fish has a slightly browned surface. Why? You decide to sprinkle the surface with a tiny bit of sugar. What happens? Why?
4. On your diagram in 1, draw the temperature in the fish as a function of distance from the center *about 15 min after* taking it off the griddle.

Viscosity

- Rank the following food items below from the least viscous to the most viscous.
Explain why you ranked them as you did.
Cream, peanut butter, water, honey

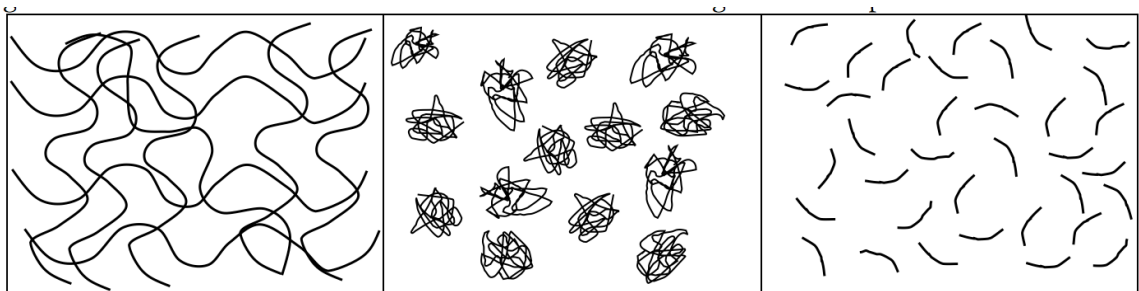
- Circle one:

Water is a	simple fluid	complex fluid
Wine is a	simple fluid	complex fluid
Mac&Cheese sauce is a	simple fluid	complex fluid

- Match the blanks. Viscosity increases when kinetic energy of molecules _____; viscosity _____ when the volume fraction decreases.

- decreases/decreases
- decreases/increases
- increases/increases
- increases/decreases

- The images below show cartoons of three sauces from a microscopic view.
Rank them from expected lowest to highest viscosity.



- What type of thickener would you expect that each image represents? How does it thicken?

Emulsions and Foams

1. You emulsify 30 mL of olive oil into one (10 ml) egg yolk to make aioli. What is occurring at a microscopic scale as you form this emulsion?
2. Why do you need to put in energy to make an emulsion?
3. What is the volume fraction of the oil in the aioli?
4. Based on your calculated volume fraction of olive oil in aioli, what is the elastic modulus of this emulsion, assuming $\sigma_{oil} = 10 \text{ mN/m}$ and the droplets $\sim 10 \text{ }\mu\text{m}$ in diameter?
5. All of these foods are common emulsifiers except:
 - a) egg yolk
 - b) garlic
 - c) mustard
 - d) brine